Stochastic Processes

Homework 2: Random Variables, Moments, and Asymptotics

Lecture 2 | 20 problems | 100 points

Assigned reading

Sheldon M. Ross, *Introduction to Probability Models*, 10th ed.

- **Primary:** Chapter 2: random variables, distributions, expectation, variance, covariance, and joint distributions.
- **Supporting:** Review the law of large numbers and central limit theorem from lecture notes and relevant sections of the text.

Instructions

Answer all 20 questions. Show your reasoning, state assumptions, and identify the probability rules or distributions used. Give exact answers when practical; round numerical probabilities to four decimal places. You may use a calculator or Python to check arithmetic, but provide mathematical derivations. For normal approximations, show the standardized expression before evaluating it. No solutions are included in this handout.

I. Random Variables and Distributions — 16 points

Problem 1. A discrete probability law **[4 points]**

A random variable X has PMF $p_X(k) = c(k+1)$ for $k \in \{0, 1, 2, 3, 4\}$, and zero otherwise.

(a) Find the normalizing constant c.

(b) Calculate $P(X \geq 3)$.

(c) Compute $E[X]$ directly from the PMF.

(d) Compute $E[X^2]$ and $\mathrm{Var}(X)$.

Problem 2. A cumulative distribution function **[4 points]**

The PMF of X is given by

x	-2	0	1	3
$P(X = x)$	0.10	0.20	0.40	0.30

(a) Write $F_X(x)$ for all real x.

(b) Find $P(-1 < X \leq 1)$.

(c) Calculate $E[X]$.

(d) Must $E[X]$ belong to the support of X? Explain.

Problem 3. A continuous density **[4 points]**

Suppose $f_X(x) = cx^2$ for $0 \leq x \leq 2$, and $f_X(x) = 0$ otherwise.

(a) Determine c.

(b) Derive $F_X(x)$ for all real x.

(c) Find $P(X > 1)$.

(d) Evaluate $E[X]$ by integration.

Problem 4. Binomial and Poisson moments **[4 points]**

Let $X \sim \text{Binomial}(20, 0.3)$ and $Y \sim \text{Poisson}(4)$.

(a) Find $E[X]$ and $\text{Var}(X)$.

(b) Find $E[Y]$ and $\text{Var}(Y)$.

(c) Compute $E[X^2]$ and $E[Y^2]$.

(d) Explain why $E[X^2]$ need not equal $(E[X])^2$.

II. Expectation and Variance 24 points

Problem 5. Moments from first principles **[6 points]**

Let X take values $-2, 0, 1, 4$ with respective probabilities $0.1, 0.3, 0.4, 0.2$.

(a) Compute $E[X]$ and $E[X^2]$ directly.

(b) Find $\text{Var}(X)$ using the second-moment identity.

(c) Find the standard deviation of X.

(d) Compute $E[(X-1)^2]$ directly and compare it with $\text{Var}(X)$.

Problem 6. Affine transformations **[6 points]**

Suppose $E[X] = 5$ and $\text{Var}(X) = 9$. Let $Y = 3X - 4$ and $Z = -2X + 7$.

(a) Find $E[Y]$ and $E[Z]$.

(b) Find $\text{Var}(Y)$ and $\text{Var}(Z)$.

(c) Calculate $E[Y^2]$.

(d) Explain why adding a constant changes the mean but not the variance.

Problem 7. Deriving variance identities **[6 points]**

Let X have finite second moment and $\mu = E[X]$.

(a) Starting from $\text{Var}(X) = E[(X-\mu)^2]$, prove $\text{Var}(X) = E[X^2] - \mu^2$.

(b) Prove $\text{Var}(aX + b) = a^2 \,\text{Var}(X)$ for constants a, b.

(c) Explain why $\text{Var}(-X) = \text{Var}(X)$.

Problem 8. Linearity without independence **[6 points]**

Three consecutive hourly arrival counts X_1, X_2, X_3 satisfy $E[X_1] = 8$, $E[X_2] = 12$, and $E[X_3] = 10$. Let $N = X_1 + X_2 + X_3$.

(a) Find $E[N]$.

(b) If each customer generates \$15 in revenue, find $E[R]$ for $R = 15N$.

(c) With a fixed cost of \$120, find $E[\Pi]$ for $\Pi = R - 120$.

(d) Does the calculation of $E[N]$ require independence? Explain.

III. Covariance and Joint Distributions

30 points

Problem 9. Covariance from a joint PMF [6 points]

For binary variables $X, Y \in \{0, 1\}$, the joint PMF is

	$Y = 0$	$Y = 1$
$X = 0$	0.20	0.30
$X = 1$	0.10	0.40

(a) Find the marginal PMFs of X and Y.

(b) Compute $E[X]$, $E[Y]$, and $E[XY]$.

(c) Calculate $\mathrm{Cov}(X, Y)$.

(d) Determine whether X and Y are independent. Justify.

Problem 10. Variance of sums and differences [6 points]

Suppose $E[X] = 4$, $E[Y] = 7$, $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 16$, and $\mathrm{Cov}(X, Y) = -3$.

(a) Find $E[2X - Y]$.

(b) Compute $\mathrm{Var}(X + Y)$.

(c) Compute $\mathrm{Var}(2X - Y)$.

(d) Compare the variance of $X + Y$ with the value under independence.

Problem 11. Correlation and its interpretation [6 points]

Let $\mathrm{Var}(X) = 25$, $\mathrm{Var}(Y) = 36$, and $\mathrm{Cov}(X, Y) = 18$.

(a) Compute the correlation coefficient $\rho_{X,Y}$.

(b) Find $\mathrm{Cov}(3X + 2, -2Y + 5)$.

(c) Find $\mathrm{Corr}(3X + 2, -2Y + 5)$.

(d) Explain the difference between zero covariance and independence.

Problem 12. A dependent random-variable construction [6 points]

Let X take values $-1, 0, 1$ with equal probability, and define $Y = X^2$.

(a) Find $E[X]$, $E[Y]$, and $E[XY]$.

(b) Compute $\mathrm{Cov}(X, Y)$.

(c) Determine whether X and Y are independent.

(d) Explain what this example shows about covariance as a measure of dependence.

Problem 13. Continuous joint density **[6 points]**

Let (X, Y) have joint density $f_{X,Y}(x, y) = c(x + y)$ for $0 \le x \le 1$ and $0 \le y \le 1$, and zero otherwise.

(a) Determine c by normalization.

(b) Derive the marginal densities f_X and f_Y.

(c) Calculate $E[X]$, $E[Y]$, and $E[XY]$.

(d) Find $\mathrm{Cov}(X, Y)$ and determine whether X, Y are independent.

IV. Asymptotics and Limit Theorems 16 points

Problem 14. Mean and variance of a sample average **[4 points]**

Let $X_1, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define $\overline{X}_n = n^{-1} \sum_{i=1}^{n} X_i$.

(a) Derive $E[\overline{X}_n]$.

(b) Derive $\mathrm{Var}(\overline{X}_n)$, explaining where independence is used.

(c) Evaluate the standard deviation when $\sigma = 6$ and $n = 144$.

Problem 15. Law of large numbers and Chebyshev's inequality **[4 points]**

Suppose $X_1, \dots, X_n$ are i.i.d. with mean 10 and variance 25.

(a) Use Chebyshev's inequality to bound $P(|\overline{X}_n - 10| \ge 1)$.

(b) Find a sufficient sample size n to make this bound at most 0.05.

(c) State the weak law of large numbers and explain its relevance here.

Problem 16. Central limit theorem for averages **[4 points]**

Independent service times $X_1, \dots, X_{100}$ are identically distributed with mean 12 minutes and standard deviation 4 minutes.

(a) Find $E[\overline{X}_{100}]$ and $\mathrm{Var}(\overline{X}_{100})$.

(b) Approximate $P(11.2 \le \overline{X}_{100} \le 12.8)$ using the CLT.

(c) Explain why the CLT is an approximation rather than an exact distributional claim here.

Problem 17. Asymptotics for a sum **[4 points]**

Let $X_1, \dots, X_{400}$ be i.i.d. with mean 3 and variance 4, and set $S_{400} = \sum_{i=1}^{400} X_i$.

(a) Find $E[S_{400}]$ and $\mathrm{Var}(S_{400})$.

(b) Approximate $P(S_{400} > 1240)$ using the CLT.

(c) Explain the difference between the scale of fluctuations of S_n and $\overline{X}_n$.

V. Synthesis and Mathematical Reasoning 14 points

Problem 18. A common shock and correlated counts [4 points]

Let U, V, W be independent with $E[U] = 2$, $E[V] = 3$, $E[W] = 4$ and variances $1, 2, 3$, respectively. Set $X = U + W$ and $Y = V + W$.

(a) Find $E[X]$, $E[Y]$, $\mathrm{Var}(X)$, and $\mathrm{Var}(Y)$.

(b) Calculate $\mathrm{Cov}(X, Y)$.

(c) Find $\mathrm{Var}(X - Y)$ and interpret the cancellation of the common shock.

Problem 19. A random sum: conditional moments [5 points]

Let $N \sim \mathrm{Poisson}(5)$. Conditional on N, let $Z_1, \ldots, Z_N$ be i.i.d. with mean 2 and variance 3, independent of N. Define $S = \sum_{i=1}^{N} Z_i$, with $S = 0$ when $N = 0$.

(a) Compute $E[S \mid N]$ and $\mathrm{Var}(S \mid N)$.

(b) Use the law of total expectation to find $E[S]$.

(c) Using $\mathrm{Var}(S) = E[\mathrm{Var}(S \mid N)] + \mathrm{Var}(E[S \mid N])$, calculate $\mathrm{Var}(S)$.

(d) Explain the two sources of randomness in S.

Problem 20. Portfolio variance and diversification [5 points]

Two random returns R_1, R_2 satisfy $E[R_1] = 0.06$, $E[R_2] = 0.10$, $\mathrm{Var}(R_1) = 0.04$, $\mathrm{Var}(R_2) = 0.09$, and $\mathrm{Cov}(R_1, R_2) = 0.03$. Define $R_p = wR_1 + (1 - w)R_2$.

(a) Derive formulas for $E[R_p]$ and $\mathrm{Var}(R_p)$ in terms of w.

(b) Calculate the portfolio mean and variance at $w = 1/2$.

(c) Find the value of $w \in [0, 1]$ that minimizes portfolio variance.

(d) Explain how covariance affects the benefits of diversification.

End of Homework 2. Show all calculations and justify any limit-theorem approximations.